



CYBERSECURITY-101 BOOTCAMP

STUDIO SHODWE

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consectetur adipiscing elit, sed do
eiusmod tempor incididunt ut labore
et dolore magna aliqua.

What is Cybersecurity?



CYBERSECURITY PROTECTS OUR DEVICES, NETWORKS, APPS, AND DATA FROM DIGITAL THREATS.

- KEEPS PRIVATE INFORMATION PRIVATE.
- PROTECTS ACCOUNTS AND FILES.
- HELPS WEBSITES AND SERVICES STAY AVAILABLE.
- REDUCES DAMAGE FROM ATTACKS.

THINK OF IT AS SECURITY FOR YOUR DIGITAL LIFE.

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03 The CIA of Cybersecurity

YOUR COLLEGE MARKS MAKE A USEFUL EXAMPLE :

Goal	Meaning	Example
Confidentiality	Only authorised people can see information	Only you and authorised staff can view your marks
Integrity	Information stays accurate	Nobody secretly changes your marks
Availability	Information works when needed	The results portal opens when you need it

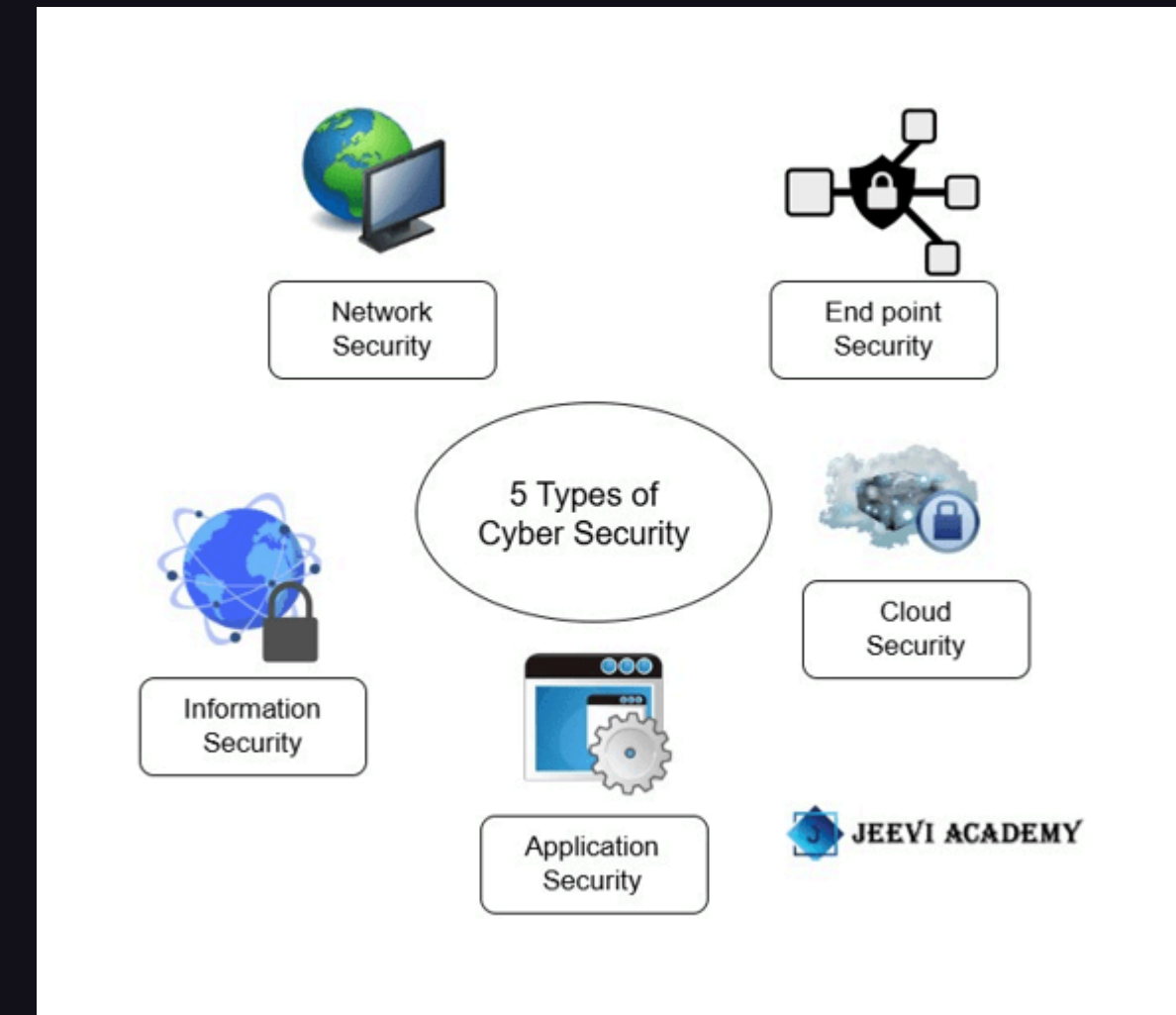


A QUICK QUESTION TO EVERYONE :

SOMEONE CHANGES YOUR MARKS WITHOUT PERMISSION.
WHICH GOAL FAILED?

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Where cybersecurity works?



NETWORK SECURITY

WHAT IT PROTECTS

Wi-Fi,
connections,
and network
traffic

APPLICATION SECURITY

WHAT IT PROTECTS

Apps,
websites,
and software

INFORMATION SECURITY

WHAT IT PROTECTS

Files and
sensitive
information

CLOUD SECURITY

WHAT IT PROTECTS

Online storage
and cloud
services

Software Devices and people need protection too

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Endpoint security	Protects laptops, phones, and other devices.
Operational security	Covers access rules, staff training, and incident handling
IoT security	Protects connected devices such as smart TVs and CCTV cameras.

Even a smart camera needs someone watching its security.



Malware and its unwanted friends

RANSOMWARE

Locks your data and demands payment for the decryption key

TROJANS

Disguises itself as legitimate software to create backdoors or steal data

SPYWARE

Secretly watches activity and collects information

ROOTKITS

Hide malicious activity and help attackers maintain access



Analogy

An unwanted guest who enters your house, watches you, and hides their footprints.



Common Attacks

Phishing:
Fake messages or websites trick you into sharing information

Ransomware:
Locks or encrypts files and demands payment.

DDoS:

Floods a service with traffic, making access difficult for real users.

analogy:

Thousands of fake orders overwhelm the canteen while you wait for one samosa.

A “free prize” message asks for your account password. Which attack does it resemble?

Changing security priorities



- **Ransomware defence:** Protecting files and preparing for recovery.
- **Cloud and IoT:** Securing online services and connected devices.
- **Zero Trust:** Verify access requests instead of automatically trusting a device because it is inside the network.
- **AI:** Helps security teams identify suspicious patterns.

Why Linux matters

- **Open source:** People can inspect and modify its code.
- **Powerful terminal:** Commands make many tasks quick.
- **Automation:** Bash and Python help repeat tasks.
- **Security tools:** Kali Linux and Parrot OS collect useful tools.
- **Logs:** Records of system activity help investigations.



A **virtual machine** gives you a practice computer inside your computer.

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Linux comes in different flavours

A distribution, or distro, packages Linux with tools and software.

Group	Examples	Common use
Debian-based	Debian, Ubuntu, Linux Mint	Everyday use and beginner learning
Red Hat-based	Fedora, RHEL, Rocky Linux	Workstations and servers
Arch-based	Arch Linux, Manjaro	Customisation and more hands-on setup
Security-focused	Kali Linux, Parrot OS	Security testing and learning



Let's Start with Learning and Implementation



Step 1: Open Chrome and login to your Github Account

Finding your way around

THE TERMINAL VERSION OF
CHECKING YOUR LOCATION AND
OPENING DOORS.

COMMAND	MEANING
<code>pwd</code>	Where am I?
<code>ls</code>	What is in this folder?
<code>cd folder</code>	Enter a folder
<code>cd ..</code>	Go up one folder

Linux filenames beginning with a dot are hidden from a normal ls listing.

Example:

```
ls -a
```

Shows all files, including hidden ones.

Example:

```
ls -la
```

Adds details such as permissions, owner, and size.

Hidden files

HIDDEN MEANS **LESS VISIBLE.**
IT DOES NOT MEAN
ENCRYPTED.

COMMAND	MEANING
mkdir test	Creates a folder called test
cp a.txt b.txt	Copies a.txt to b.txt
mv old.txt new.txt	Renames the file
mv file.txt test/	Moves the file into test

Creating, copying, and moving

Memory trick:

cp makes another copy. mv changes the name or location.

Reading and editing files

Vim survival kit:

- Press I to start typing.
- Press Esc to leave typing mode.
- Type :wq and press Enter to save and exit.

COMMAND	MEANING
cat file.txt	Displays the file's contents
nano file.txt	Opens a simple text editor
vim file.txt	Opens an editor with different modes
cat file.txt	Displays the file's contents

COMMAND	MEANING
rm file.txt	Deletes a file
rm -r test	Deletes a folder and its contents
rm -rf test	Adds force, skipping many prompts and errors

Deleting files carefully

THESE COMMANDS USUALLY **BYPASS** THE RECYCLE BIN.

CHECK YOUR LOCATION WITH PWD BEFORE DELETING.

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Number	Permission
4	Read
2	Write
1	Execute

drwxrwxrwx

d = Directory
r = Read
w = Write
x = Execute

chmod 777

rwx | rwx | rwx

Owner | Group | Others

7	rwx	111
6	rw-	110
5	r-x	101
4	r--	100
3	-wx	011
2	-w-	010
1	--x	001
0	---	000

Add them together:
7 = read + write + execute.

Example:

chmod 755 script.sh

Update system to
save changes:
sudo apt update

Owner gets
7: read, write, execute.

Group and others get
5: read and execute.

Runs the command with administrator
privileges and refreshes package
information on Debian-based
systems.

sudo gives a command extra
power.

GREP **SEARCHES TEXT**, LIKE CTRL+F IN THE TERMINAL.

Example:

```
grep "password" file.txt
```

FINDS LINES CONTAINING PASSWORD.

Example:

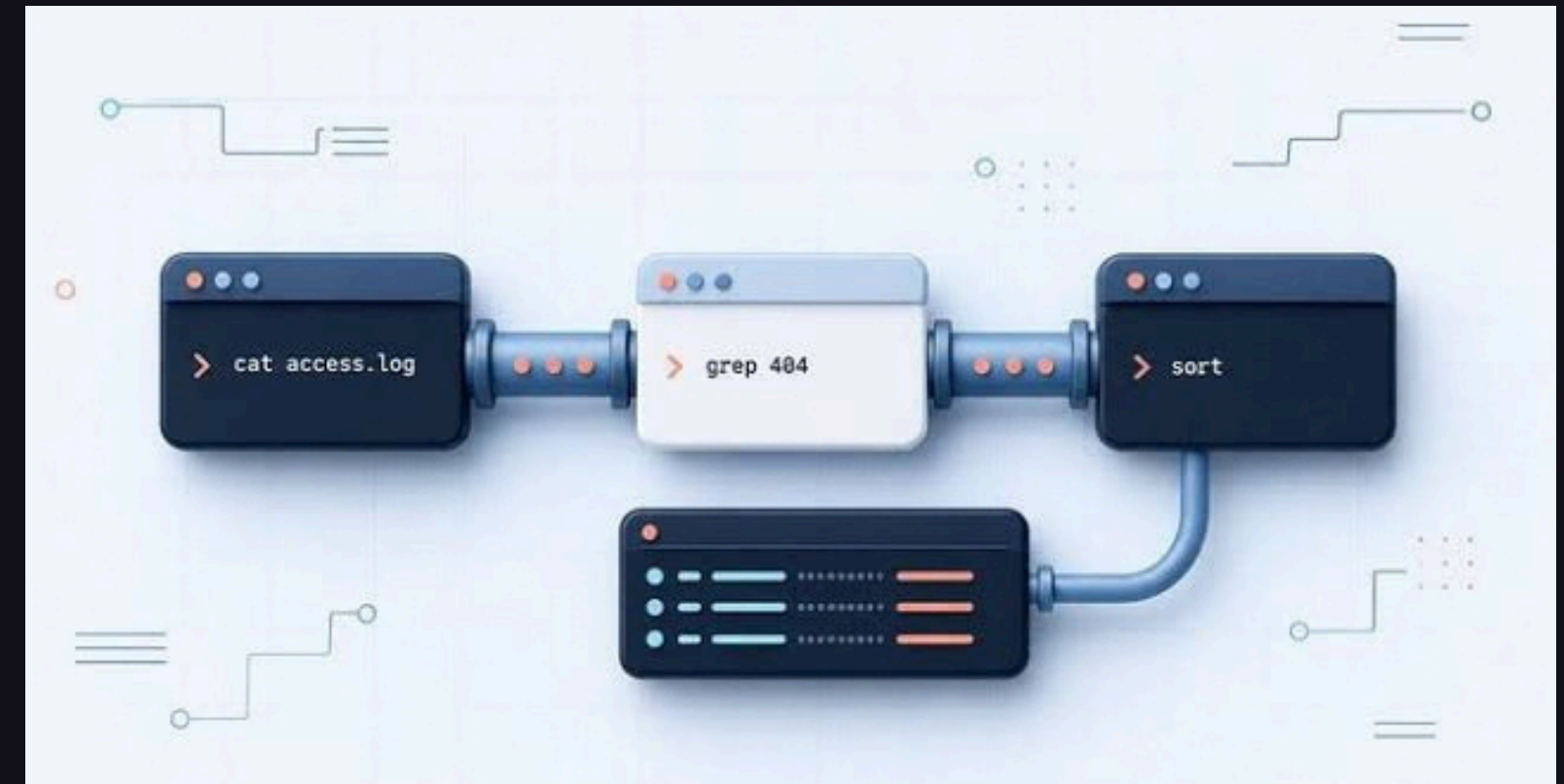
```
grep -i "error" log.txt
```

Matches error, ERROR, and other capitalisation.

Example:

```
ps aux | grep apache
```

grep and pipes 18



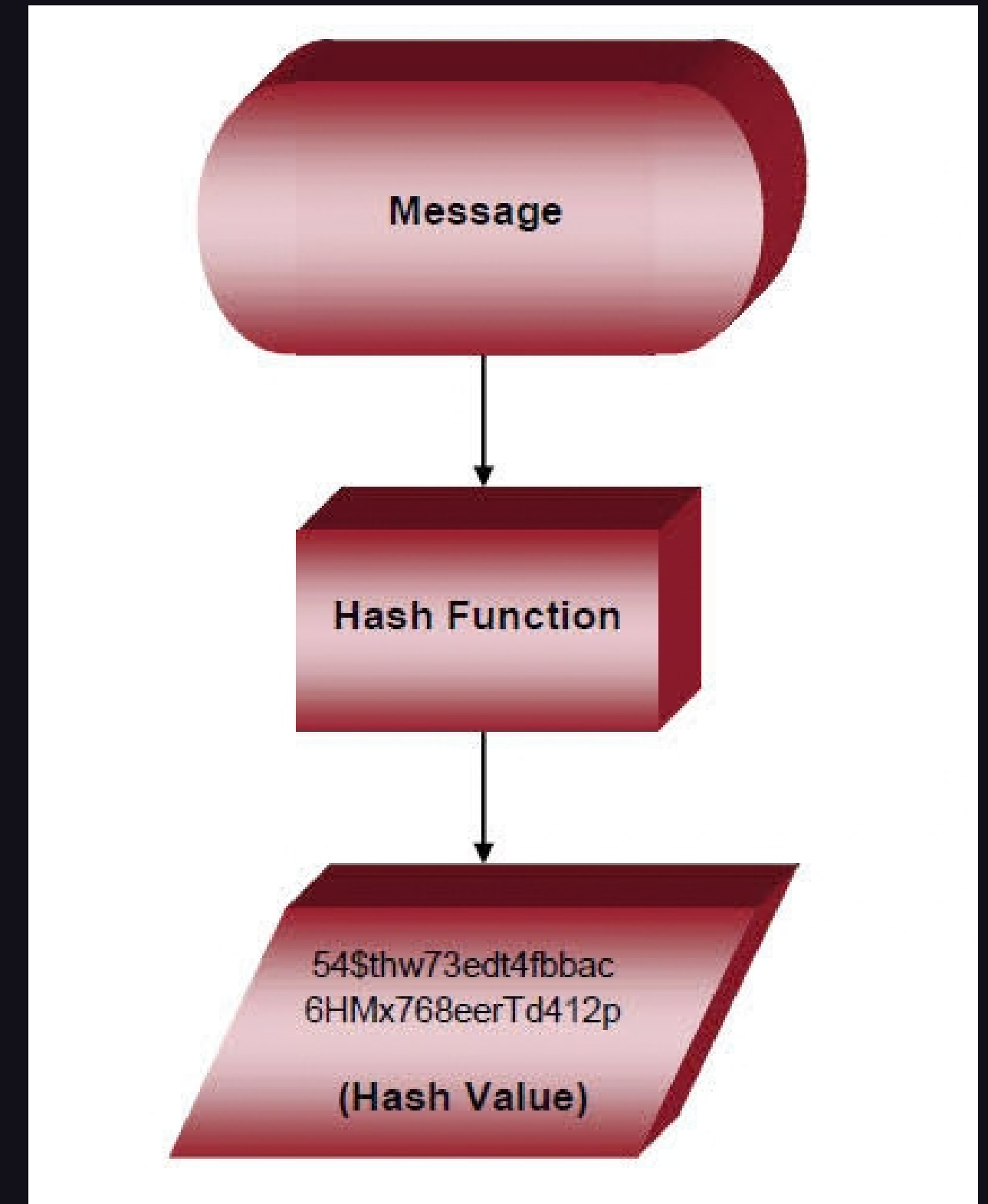
SEARCHES THE **RUNNING-PROCESS** LIST FOR APACHE.

THE **PIPE | PASSES** ONE COMMAND'S OUTPUT **TO THE NEXT COMMAND**.

Hashing

- A hash **converts data into a fingerprint** of a fixed length.
- The **same input produces the same hash**.
- A **tiny change** produces a **very different-looking hash**.
- Hashes help check whether **files have changed**.
- There is **no decryption operation** for a hash.

However, attackers can **guess an input** and compare its hash.



SHA-256 produces 64 hexadecimal characters.

One letter changes the fingerprint

```
printf 'hello' | sha256sum  
printf 'Hello' | sha256sum
```

You can also demonstrate an older hash:

```
printf 'hello' | md5sum
```

Input	Beginning of SHA-256 hash
hello	2cf24dba...
Hello	185f8db3...

MD5 and SHA-1 have known weaknesses and should not serve as modern security checks.

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Base64 changes the format

Base64 represents data using a **set of printable characters**.

ENCODE:

```
printf 'hello' | base64
```

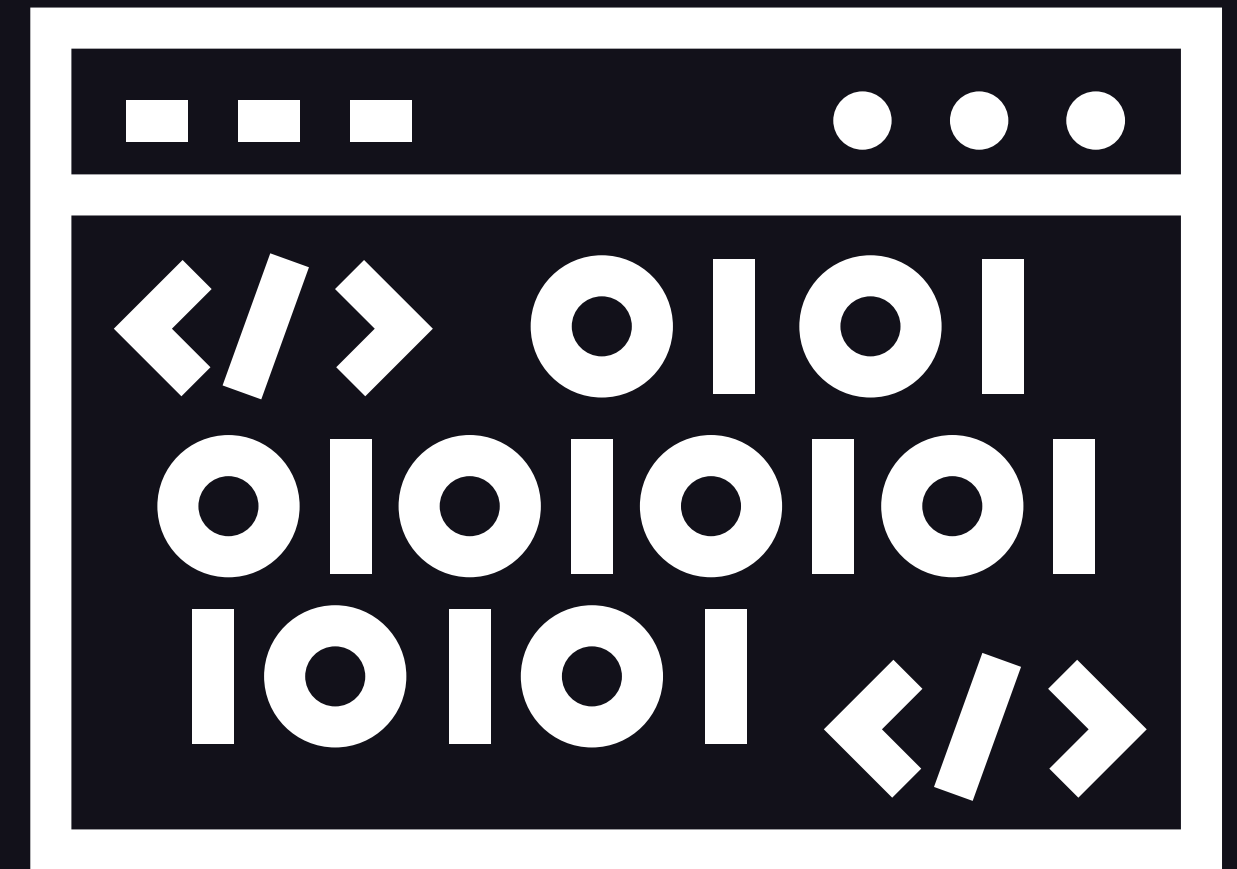
OUTPUT:

```
aGVsbG8=
```

ENCODE:

```
printf 'aGVsbG8=' | base64 -d
```

Result: hello



Anyone can decode Base64 without a secret key. It **does not provide secrecy**.

Binwalk, the file detective

A file can contain **other files** or **embedded data**.

Example:

A picture might contain an appended ZIP archive.

```
binwalk image.jpg
```

```
binwalk -e image.jpg
```

Attempts to **extract supported embedded content**.

Useful for CTFs, digital forensics, and firmware analysis.

strings, readable clues inside files

Show the readable text:

```
strings -a mystery.bin
```

Search that text for a flag:

```
strings -a mystery.bin | grep -i "flag"
```

strings finds readable text inside files, including images and programs.

- Can reveal URLs, messages, or CTF flags.
- Normally looks for sequences of at least 4 printable characters.
- -a tells it to scan the whole file.

Example clue:
FLAG{hiding_in_plain_sight}

Steganography, a secret inside a picture

Steganography hides information inside another file while keeping its ordinary appearance.

- A normal-looking photo might carry a secret message.
- Images and audio can act as hiding places.
- In CTFs, the hidden content might contain a flag.

Demo with steghide: Start with photo.jpg and secret.txt.

```
steghide embed -cf photo.jpg -ef secret.txt -sf puzzle.jpg
```

Creates puzzle.jpg containing the secret.

```
steghide extract -sf puzzle.jpg
```

Memory trick: Encryption protects a message's meaning. Steganography conceals its presence.

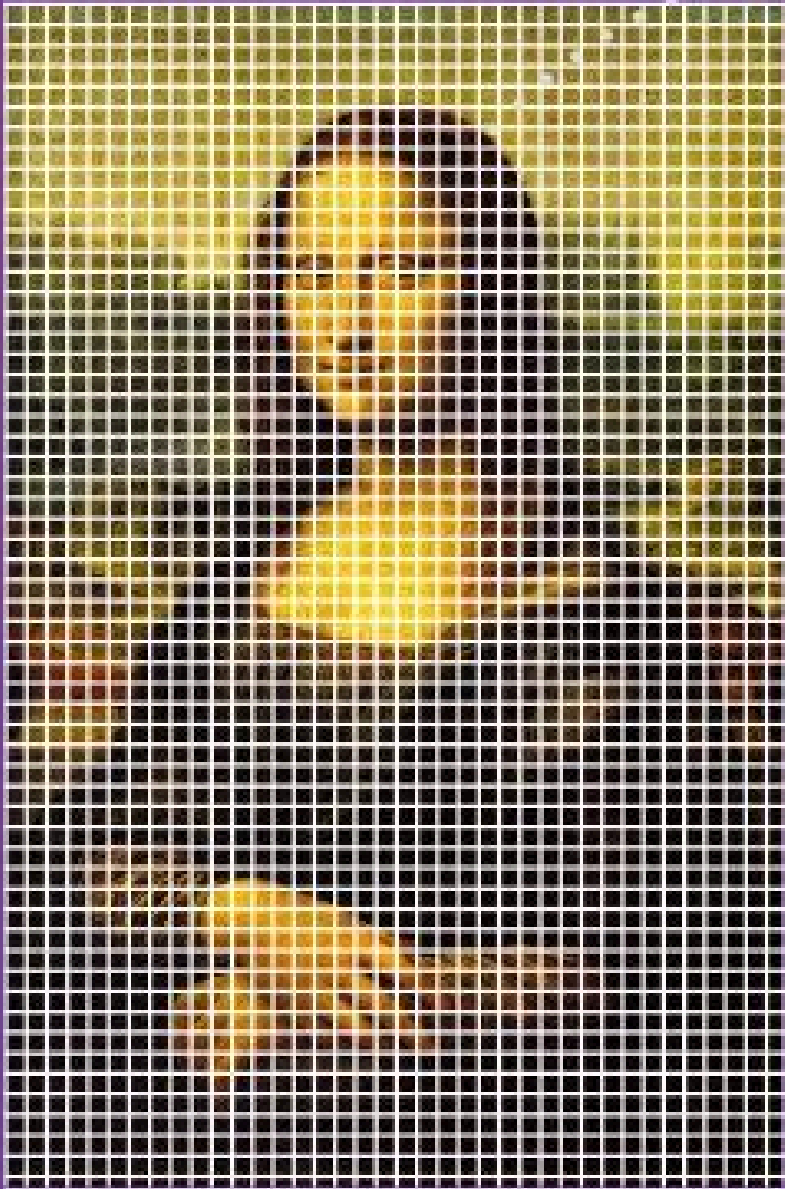
Extracts the message using the same passphrase. This extraction method works for data embedded with steghide

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Steganography, a secret inside a picture

Digital Steganography

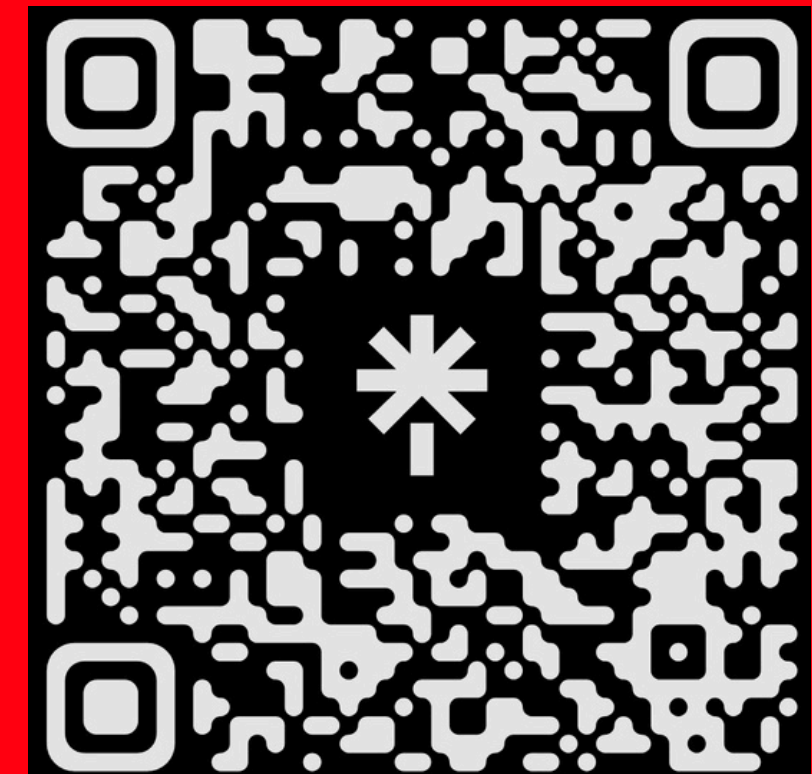
LSB IN IMAGES



144	141	81
10010000	10001101	01010001
Hidden message: 101001...		
145	140	81
1001000 1	1000110 0	0101000 1
146	142	81
100100 10	100011 10	010100 01

THANK YOU!

AND STAY TUNED FOR DAY-2



SCAN TO CONNECT ON SOCIALS

PROTOCOL CYBER-CLUB